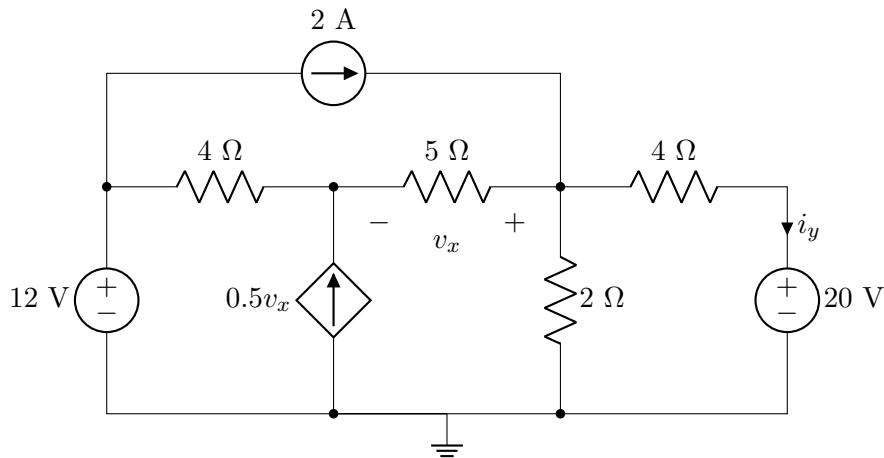


Question 1 of 3 [20 marks]

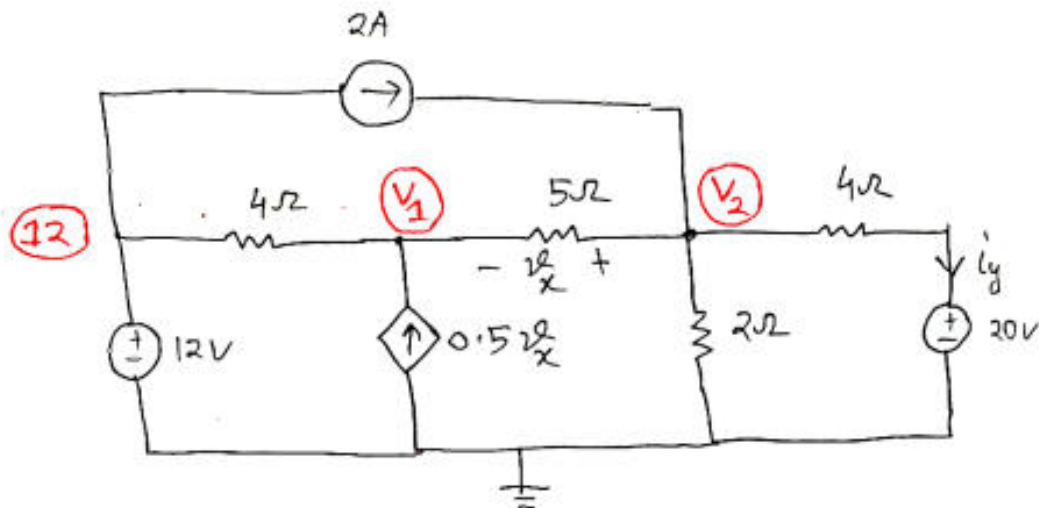
Set: 1



- **Apply Nodal/Mesh Analysis to identify** the values of the node voltages/mesh currents.
(Equations must be highlighted using boxes).
(18 marks) [CO2, CO4]
- **What** is the value of v_x and i_y? (with appropriate ± sign)?
(2 marks) [CO1, CO2]

set 1

Q1 Nodal



KCL at node 1,

$$0.5 \frac{V_x}{2} = \frac{V_1 - 12}{4} + \frac{V_1 - V_2}{5}$$

$$\Rightarrow 0.5(V_2 - V_1) = \frac{V_1}{4} - 3 + \frac{V_1}{5} - \frac{V_2}{5} \quad [V_x = V_2 - V_1]$$

$$\Rightarrow 10V_2 - 10V_1 = 5V_1 - 60 + 4V_1 - 4V_2$$

$$\Rightarrow \boxed{19V_1 - 14V_2 = 60} \quad \text{--- (i)}$$

KCL at node 2,

$$2 = \frac{V_2 - V_1}{5} + \frac{V_2}{2} + \frac{V_2 - 20}{4}$$

$$\Rightarrow 40 = 4V_2 - 4V_1 + 10V_2 + 5V_2 - 100$$

$$\Rightarrow \boxed{4V_1 - 19V_2 = -140} \quad \text{--- (ii)}$$

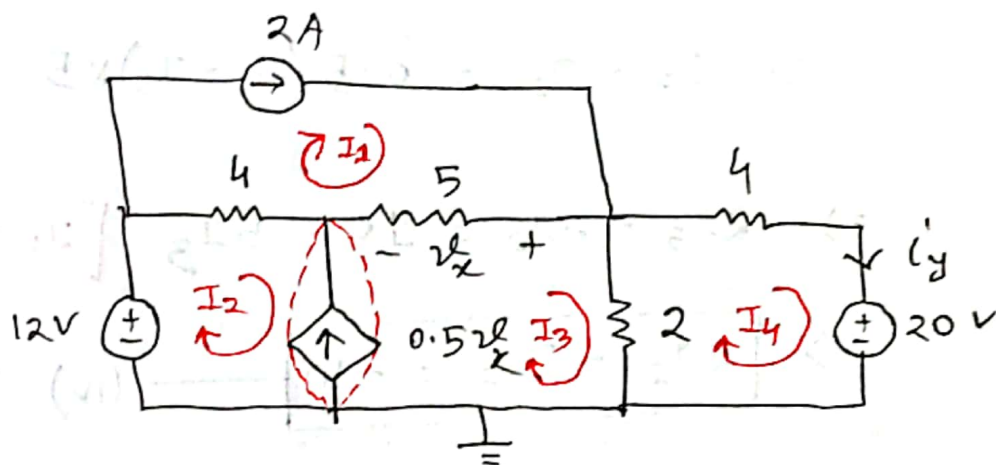
solving (i) & (ii) ...

$$V_1 = 10.16 \text{ (V)} ; V_2 = 9.51 \text{ (V)}$$

$$V_x = V_2 - V_1 = -0.65 \text{ (V)}$$

$$i_y = \frac{V_2 - 20}{4} = -2.62 \text{ (A)}$$

Mesh



$$\boxed{I_1 = 2} \text{ — (i)}$$

KVL at loops 2, & 3 (supermesh),

$$-12 + 4(I_2 - I_1) + 5(I_3 - I_1) + 2(I_3 - I_4) = 0$$

$$\Rightarrow -12 + 4I_2 - 8 + 5I_3 - 10 + 2I_3 - 2I_4 = 0 \quad [I_1 = 2]$$

$$\Rightarrow \boxed{4I_2 + 7I_3 - 2I_4 = 30} \text{ — (ii)}$$

KVL at loop 3,

$$2(I_4 - I_3) + 4I_4 + 20 = 0$$

$$\Rightarrow \boxed{2I_3 - 6I_4 = 20} \text{ --- (iii)}$$

At the supernode mesh,

$$I_3 - I_2 = 0.5 v_x$$

$$\Rightarrow I_3 - I_2 = 0.5 (I_1 - I_3) \times 5 \quad [v_x = 5(I_1 - I_3)]$$

$$\Rightarrow 2I_3 - 2I_2 = 10 - 5I_3 \quad [I_1 = 2]$$

$$\Rightarrow \boxed{2I_2 - 7I_3 = -10} \text{ --- (iv)}$$

Solving (ii), (iii), & (iv) ...

$$I_2 = 2.46 \text{ (A)}$$

$$I_3 = 2.13 \text{ (A)}$$

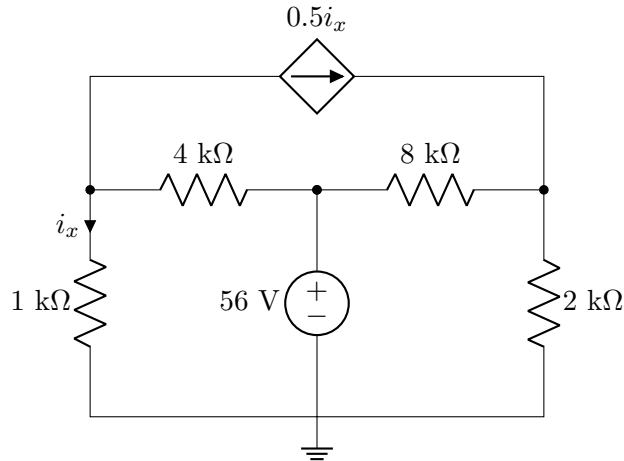
$$I_4 = -2.62 \text{ (A)}$$

$$v_x = 5(I_1 - I_3) = -0.65 \text{ (V)}$$

$$i_y = I_4 = -2.62 \text{ (A)}$$

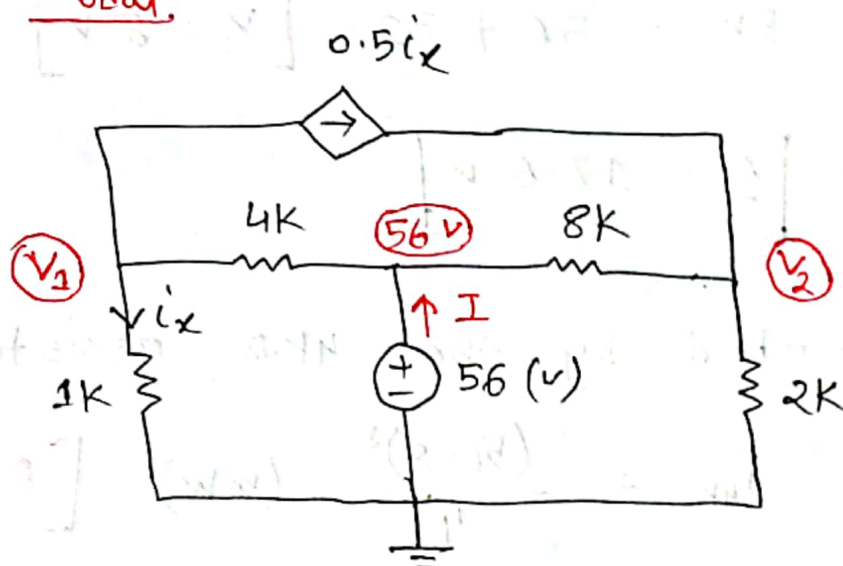
Question 2 of 3 [20 marks]

Set: 1



- **Apply Nodal/Mesh Analysis** to **identify** the values of the node voltages/mesh currents.
(Equations must be highlighted using boxes).
(14 marks) [CO2, CO4]
- **What** is the power absorbed by the 4 kΩ resistor? (with appropriate $\pm$ sign)?
(2 marks) [CO1, CO2]
- **What** is the power supplied/absorbed by the voltage source? (with appropriate $\pm$ sign)?
(4 marks) [CO1, CO2]

Q2 Nodal



KCL at node 1,

$$0.5 i_x + \frac{V_1}{1} + \frac{V_1 - 56}{4} = 0$$

$$\Rightarrow 0.5 \frac{V_1}{1} + \frac{V_1}{1} + \frac{V_1 - 56}{4} = 0 \quad \left[i_x = \frac{V_1}{1} \right]$$

$$\Rightarrow 6V_1 + V_1 - 56 = 0$$

$$\Rightarrow 7V_1 = 56$$

$$\Rightarrow \boxed{V_1 = 8(V)}$$

KCL at node 2,

$$0.5 i_x = \frac{V_2}{2} + \frac{V_2 - 56}{8}$$

$$\Rightarrow 0.5 \frac{V_1}{1} = \frac{V_2}{2} + \frac{V_2}{8} - 7 \quad \left[i_x = \frac{V_1}{1} \right]$$

$$\Rightarrow 4V_1 = 4V_2 + V_2 - 56$$

$$\Rightarrow 5V_2 = 56 + 32 \quad [V_1 = 8V]$$

$$\Rightarrow \boxed{V_2 = 17.6V}$$

Power absorbed by the $4k\Omega$ resistor,

$$P_{4K} = \frac{(56-8)^2}{4} \text{ (mW)} \quad \left[P = \frac{V^2}{R} \right]$$

$$\boxed{P_{4K} = 576 \text{ mW}}$$

Let the $56V$ source is supplying current I as shown in the figure above,

KCL at the middle node,

$$I = \frac{56 - V_1}{4} + \frac{56 - V_2}{8}$$

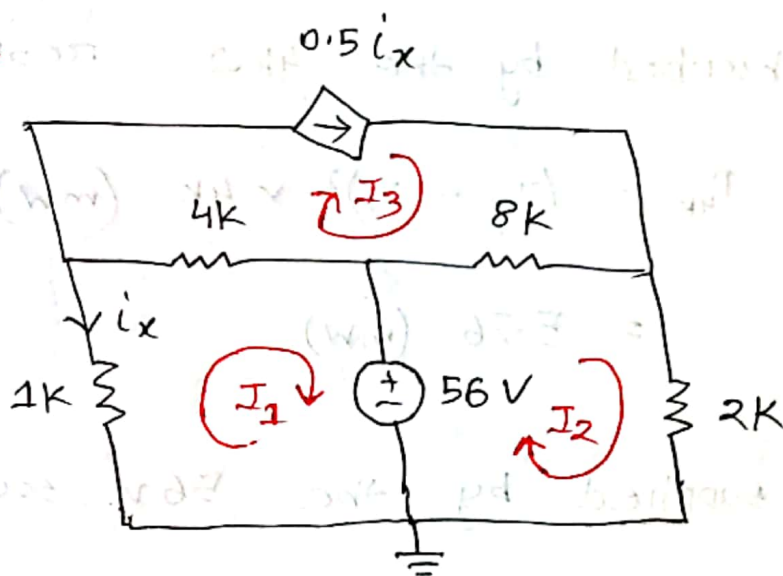
$$\Rightarrow I = 16.8 \text{ (mA)} \quad [V_1 = 8V ; V_2 = 17.6V]$$

so, power supplied by the voltage source

$$= -56 \times 16.8 \text{ (mW)}$$

$$= -940.8 \text{ (mW)}$$

Mesh



KVL at mesh 1,

$$I_1 + 4(I_1 - I_3) + 56 = 0$$

$$\Rightarrow \boxed{5I_1 - 4I_3 = -56} \quad \text{--- (i)}$$

KVL at mesh 2,

$$-56 + 8(I_2 - I_3) + 2I_2 = 0$$

$$\Rightarrow \boxed{10I_2 - 8I_3 = 56} \quad \text{--- (ii)}$$

At mesh 3,

$$I_3 = 0.5 i_x = -0.5 I_1 \quad [i_x = -I_1]$$

$$\Rightarrow \boxed{I_1 + 2I_3 = 0} \quad \text{--- (iii)}$$

Solving (i), (ii), & (iii) ...

$$I_1 = -8 \text{ (mA)} ; I_2 = 8.8 \text{ (mA)} ; I_3 = 4 \text{ (mA)}$$

Power absorbed by the $4k\Omega$ resistor,

$$\begin{aligned} P_{4k} &= (I_3 - I_1)^2 \times 4k \text{ (mW)} \\ &= 576 \text{ (mW)}. \end{aligned}$$

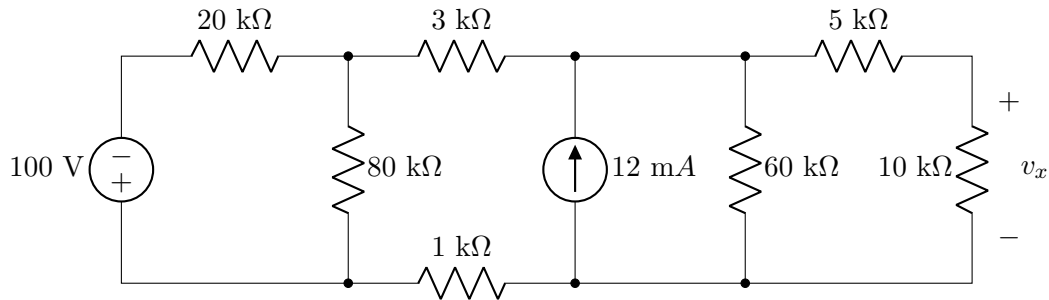
Current supplied by the $56V$ source

$$\begin{aligned} &= I_2 - I_1 \\ &= 8.8 - (-8) \\ &= 16.8 \text{ (mA)} \end{aligned}$$

Power supplied by the voltage source,

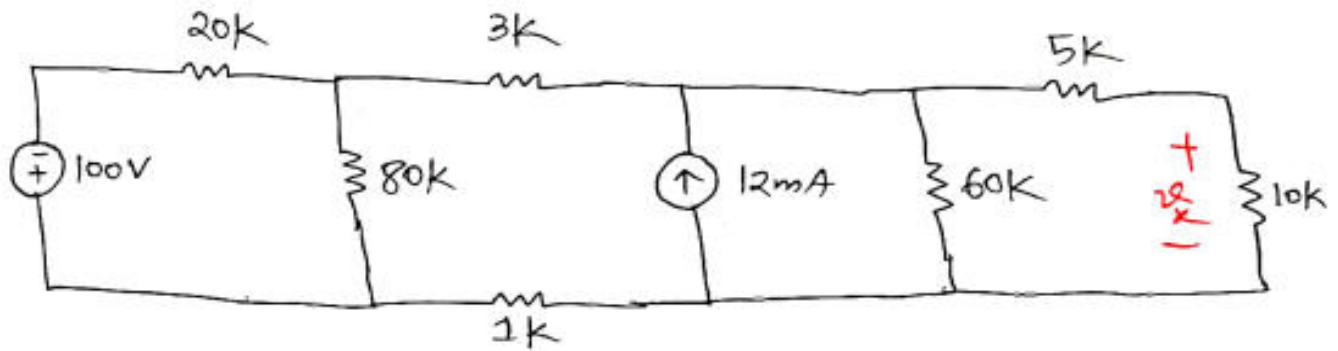
$$\begin{aligned} P &= -56 \times 16.8 \text{ (mW)} \\ &= -940.8 \text{ (mW)} \end{aligned}$$

Question 3 of 3 [15 marks]

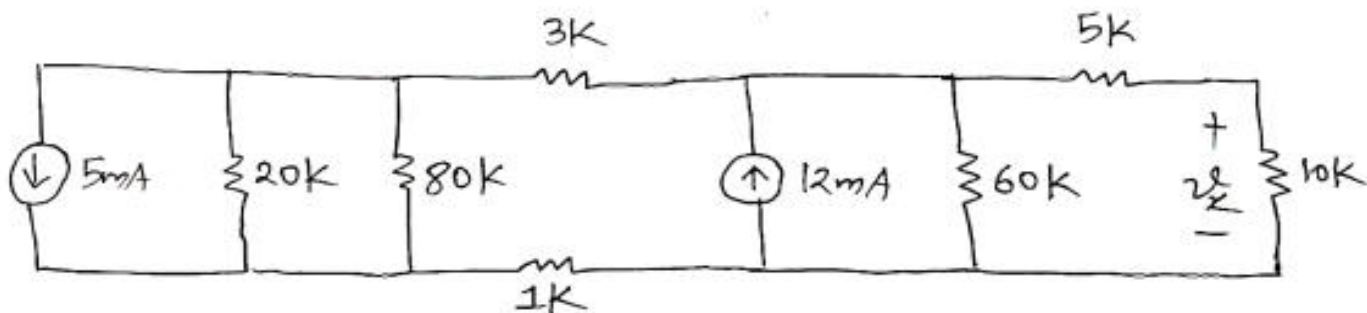


- **Determine** the voltage v_x across the 10 kΩ resistor by performing a succession of appropriate **Source Transformations**.
(15 marks) [CO2, CO4]

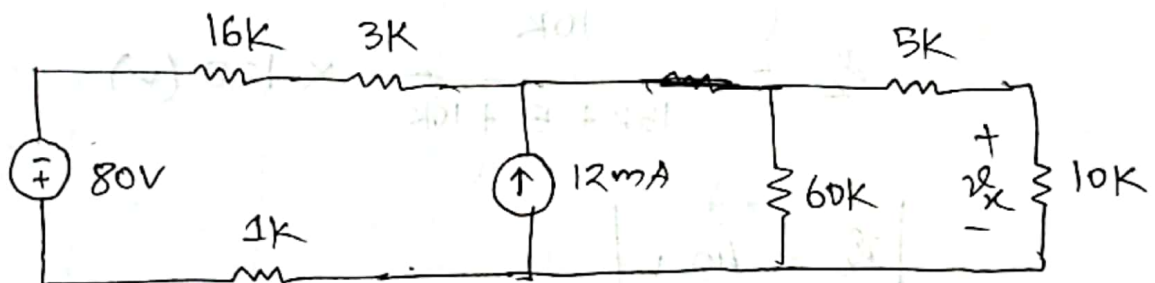
Q3



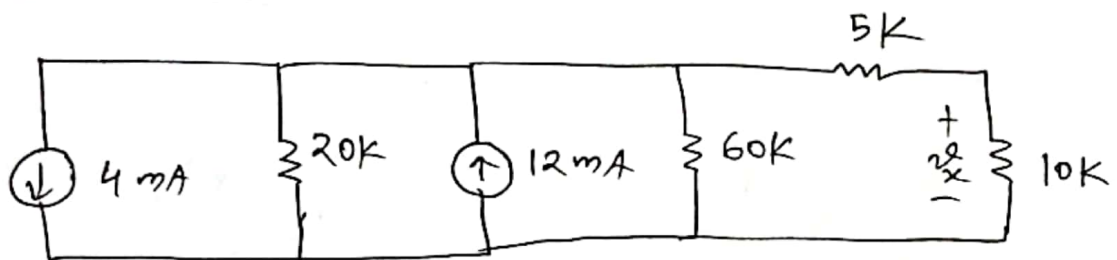
Step 1 $(100V + 20k) \rightarrow (5mA \parallel 20k)$



Step 2 & 3 $(20k \parallel 80k = 16k)$; $(5mA \parallel 16k) \rightarrow (80V + 16k)$



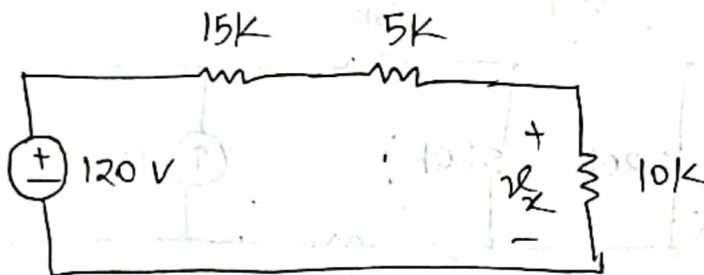
Step 4 & 5 $(16k + 3k + 1k = 20k)$; $(80V + 20k) \rightarrow (4mA \parallel 20k)$



steps 6 & 7 $(20k \parallel 60k = 15k)$; $(12mA - 4mA = 8mA)$



step 8 $(8mA \parallel 15k) \longrightarrow (120V + 15k)$



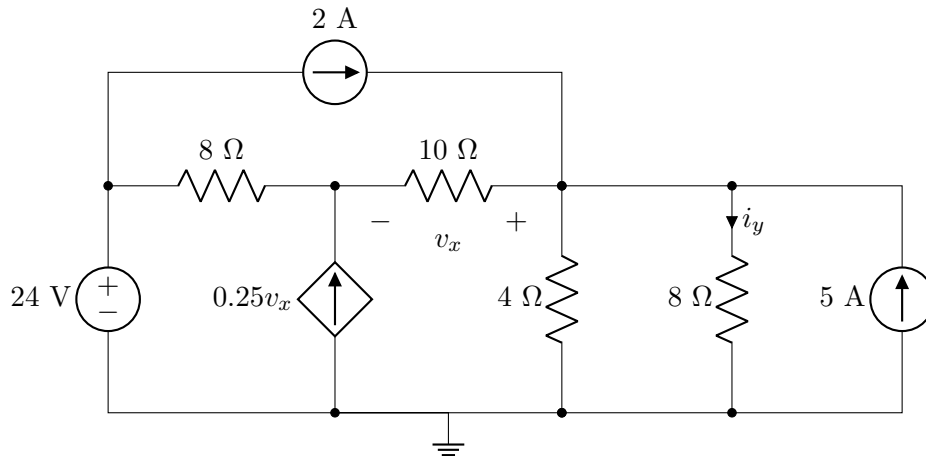
step 9 (voltage division)

$$V_x = \frac{10k}{15k + 5k + 10k} \times 120 (V)$$

$V_x = 40 V$

Question 1 of 3 [20 marks]

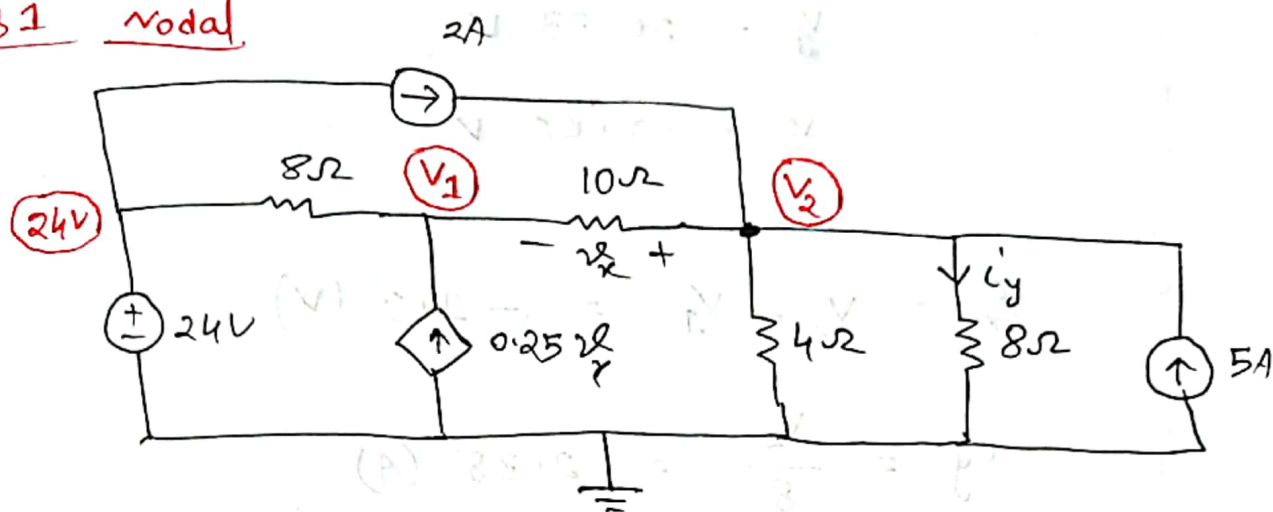
Set: 2



- **Apply Nodal/Mesh Analysis** to **identify** the values of the node voltages/mesh currents.
(Equations must be highlighted using boxes).
(18 marks) [CO2, CO4]
- **What** is the value of v_x and i_y ? (with appropriate $\pm$ sign)?
(2 marks) [CO1, CO2]

Set 2

Q1 Nodal



KCL at node 1,

$$0.25 V_x = \frac{V_1 - 24}{8} + \frac{V_1 - V_2}{10}$$

$$\Rightarrow 0.25 (V_2 - V_1) = \frac{V_1}{8} - 3 + \frac{V_1}{10} - \frac{V_2}{10} \quad [V_x = V_2 - V_1]$$

$$\Rightarrow 10V_2 - 10V_1 = 5V_1 - 120 + 4V_1 - 4V_2$$

$$\Rightarrow \boxed{19V_1 - 14V_2 = 120} \quad \text{--- (i)}$$

KCL at node 2,

$$2 + 5 = \frac{V_2 - V_1}{10} + \frac{V_2}{4} + \frac{V_2}{8}$$

$$\Rightarrow 280 = 4V_2 - 4V_1 + 10V_2 + 5V_2$$

$$\Rightarrow \boxed{4V_1 - 19V_2 = -280} \quad \text{--- (ii)}$$

solving (i) & (ii)

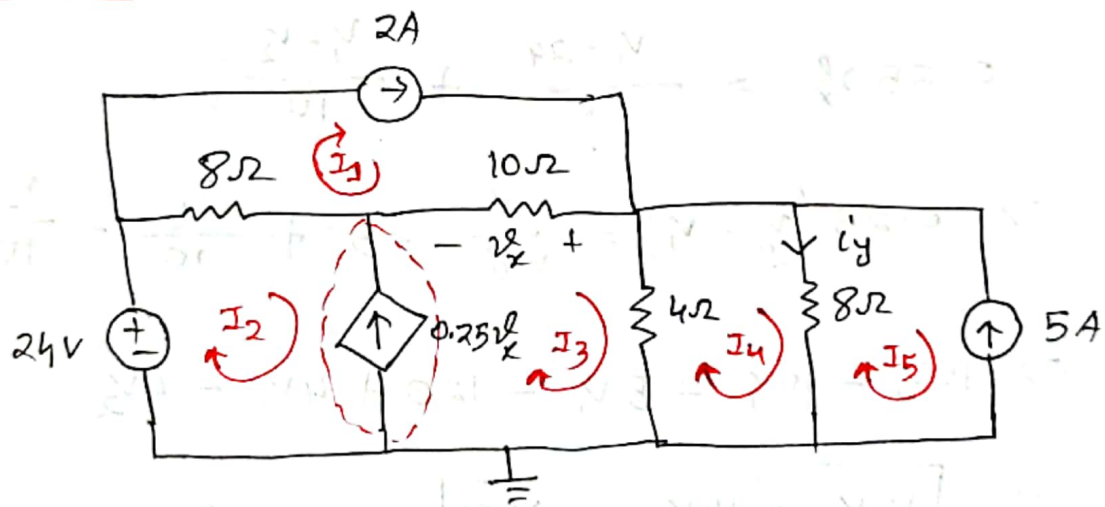
$$V_1 = 20.33 \text{ V}$$

$$V_2 = 19.02 \text{ V}$$

$$v_x = V_2 - V_1 = -1.3 \text{ (V)}$$

$$i_y = \frac{V_2}{8} = 2.38 \text{ (A)}$$

Mesh



$$I_1 = 2 \text{ A}$$

$$I_5 = -5 \text{ A}$$

KVL at mesh 2 & 3 (supermesh),

$$-24 + 8(I_2 - I_1) + 10(I_3 - I_1) + 4(I_3 - I_4) = 0$$

$$\Rightarrow -24 + 8I_2 - 16 + 10I_3 - 20 + 4I_3 - 4I_4 = 0 \quad [I_1 = 2 \text{ A}]$$

$$\Rightarrow \boxed{8I_2 + 14I_3 - 4I_4 = 60} \quad (i)$$

KVL at mesh 4,

$$4(I_4 - I_3) + 8(I_4 - I_5) = 0$$

$$\Rightarrow 4I_4 - 4I_3 + 8I_4 + 40 = 0 \quad [I_5 = -5]$$

$$\Rightarrow \boxed{4I_3 - 12I_4 = 40} \quad (ii)$$

At the supermesh,

$$I_3 - I_2 = 0.25 V_x$$

$$\Rightarrow I_3 - I_2 = 0.25 \times 10 (I_1 - I_3) \quad [V_x = 10(I_1 - I_3)]$$

$$\Rightarrow 2I_3 - 2I_2 = 5I_1 - 5I_3$$

$$\Rightarrow \boxed{-2I_2 + 7I_3 = 10} \quad (iii) \quad [I_1 = 2A]$$

Solving (i), (ii), & (iii) ...

$$I_2 = 2.46 A$$

$$I_3 = 2.13 A$$

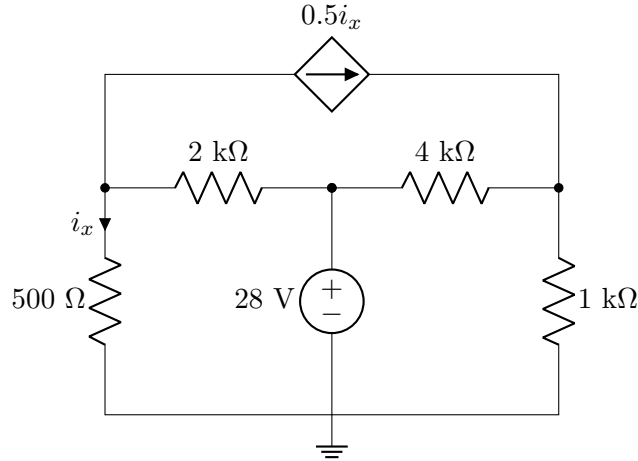
$$I_4 = -2.62 A$$

$$\begin{aligned}
 v_x &= 10 (I_1 - I_3) \\
 &= 10 (2 - 2.13) = 1.3 \text{ V}
 \end{aligned}$$

$$\begin{aligned}
 i_y &= I_4 - I_5 \\
 &= -2.62 - (-5) \\
 &= 2.38 \text{ A}
 \end{aligned}$$

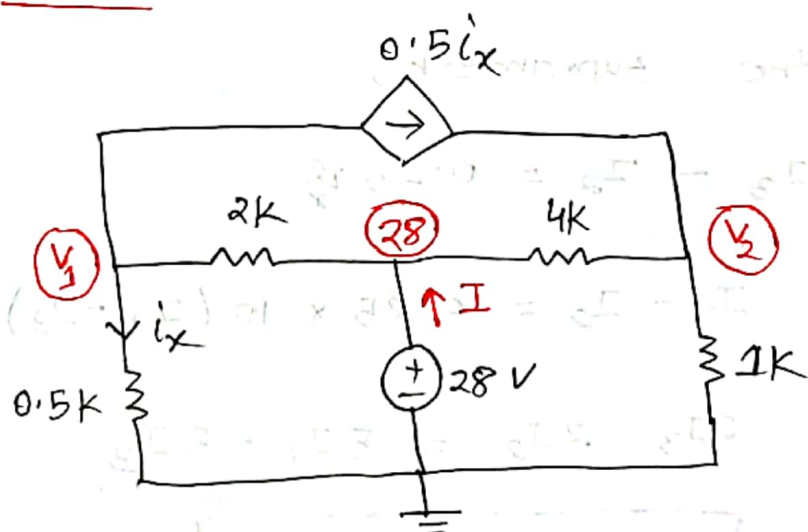
Question 2 of 3 [20 marks]

Set: 2



- **Apply Nodal/Mesh Analysis to identify** the values of the node voltages/mesh currents.
(Equations must be highlighted using boxes).
(14 marks) [CO2, CO4]
- **What** is the power absorbed by the 4 kΩ resistor? (with appropriate $\pm$ sign)?
(2 marks) [CO1, CO2]
- **What** is the power supplied/absorbed by the voltage source? (with appropriate $\pm$ sign)?
(4 marks) [CO1, CO2]

Q2 Nodal



KCL at node 1,

$$0.5i_x + \frac{V_1 - 28}{2} + \frac{V_1}{0.5} = 0$$

$$\Rightarrow 0.5 \frac{V_1}{0.5} + \frac{V_1}{2} - 14 + \frac{V_1}{0.5} = 0 \quad \left[i_x = \frac{V_1}{0.5} \right]$$

$$\Rightarrow 2V_1 + V_1 - 28 + 4V_1 = 0$$

$$\Rightarrow \boxed{V_1 = 4V}$$

KCL at node 2,

$$0.5i_x = \frac{V_2 - 28}{4} + \frac{V_2}{1}$$

$$\Rightarrow 0.5 \frac{V_1}{0.5} = \frac{V_2}{4} - 7 + V_2 \quad \left[i_x = \frac{V_1}{0.5} \right]$$

$$\Rightarrow 4V_1 = V_2 - 28 + 4V_2$$

$$\Rightarrow 5V_2 = 28 + 16 \quad \left[V_1 = 4V \right]$$

$$\Rightarrow \boxed{V_2 = 8.8V}$$

Power absorbed by the $4k\Omega$ resistor,

$$P_{4k} = \frac{(28 - V_2)^2}{4k} \text{ (mW)} \quad \left[P = \frac{V^2}{R} \right]$$

$$= 92.16 \text{ (mW)}$$

Let the current supplied by the 28V source is I as shown in the figure above,

KCL at the middle node,

$$I = \frac{28 - V_1}{2} + \frac{28 - V_2}{4}$$

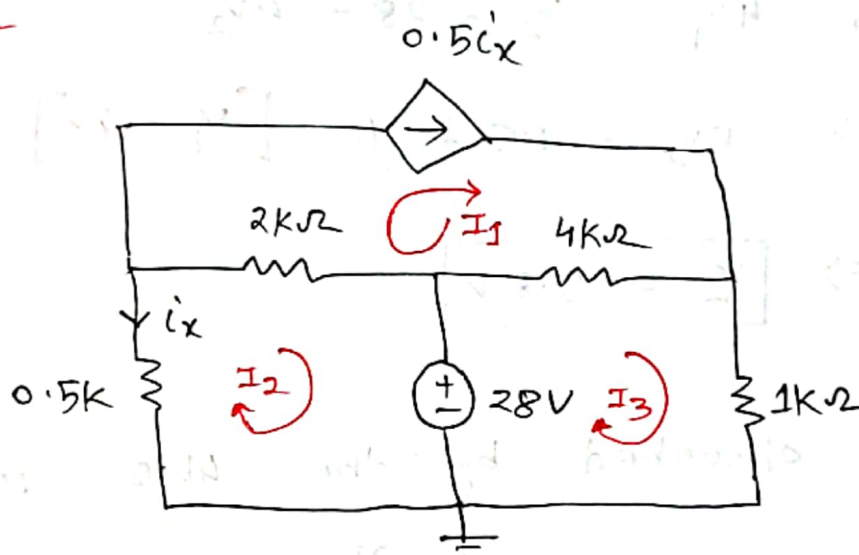
$$\Rightarrow I = 16.8 \text{ mA}$$

so, power supplied by the voltage source,

$$= -28 \times 16.8 \text{ (mW)}$$

$$= -470.4 \text{ (mW)}$$

Mesh



$$I_1 = 0.5 i_x = -0.5 I_2 \quad [i_x = -I_2]$$

$$\Rightarrow \boxed{2I_1 + I_2 = 0} \quad (i)$$

KVL at mesh 2,

$$0.5 I_2 + 2(I_2 - I_1) + 28 = 0$$

$$\Rightarrow I_2 + 4I_2 - 4I_1 + \cancel{28}^{56} = 0$$

$$\Rightarrow \boxed{4I_1 - 5I_2 = 56} \quad \text{--- (ii)}$$

KVL at mesh 3,

$$-28 + 4(I_3 - I_1) + 1I_3 = 0$$

$$\Rightarrow -28 + 4I_3 - 4I_1 + I_3 = 0$$

$$\Rightarrow \boxed{4I_1 - 5I_3 = -28} \quad \text{--- (iii)}$$

solving (i), (ii), & (iii) ...

$$I_1 = 4 \text{ mA}$$

$$I_2 = -8 \text{ mA}$$

$$I_3 = 8.8 \text{ mA}$$

Power absorbed by the $4k\Omega$ resistor,

$$P_{4k} = (I_3 - I_1)^2 \times 4k \text{ (mW)} \quad [P = I^2 R]$$

$$= 92.16 \text{ (mW)}$$

Current supplied by the 28V source

$$= I_3 - I_2 = 8.8 - (-8) = 16.8 \text{ mA}$$

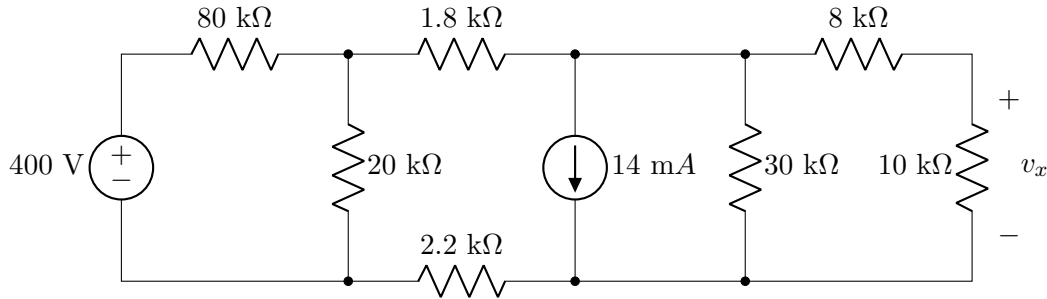
Power supplied by the voltage source

$$= -28 \times 16.8 \text{ (mW)}$$

$$= -470.4 \text{ (mW)}$$

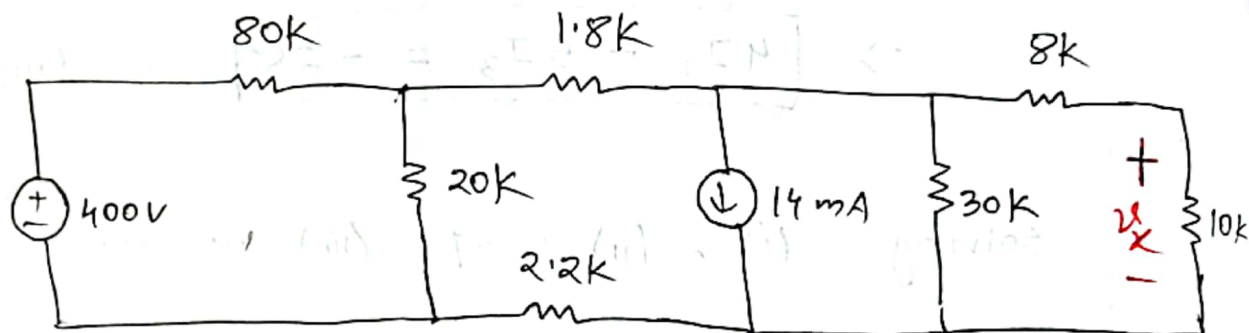
Question 3 of 3 [15 marks]

Set: 2

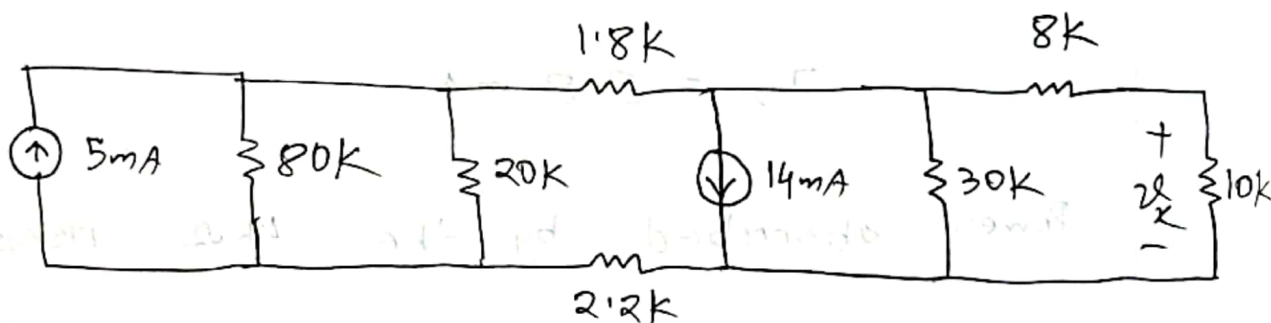


- **Determine** the voltage v_x across the 10 kΩ resistor by performing a succession of appropriate **Source Transformations**.
(15 marks) [CO2, CO4]

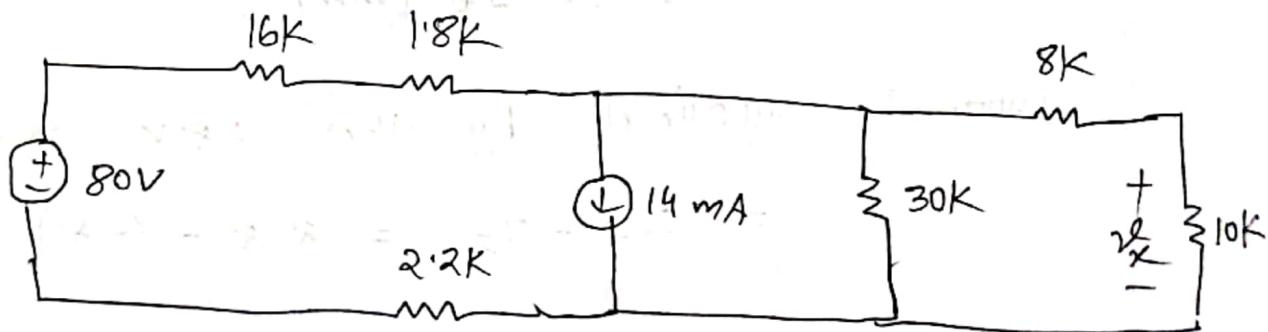
83



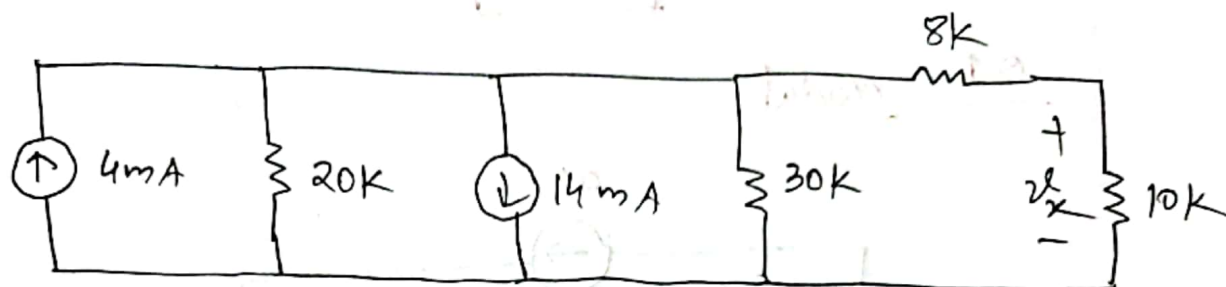
step 1 $(400V + 80k) \longrightarrow (5mA \parallel 80k)$



steps 2 & 3 $(80k \parallel 20k) = 16k$; $(5mA \parallel 16k) \longrightarrow (80V + 16k)$



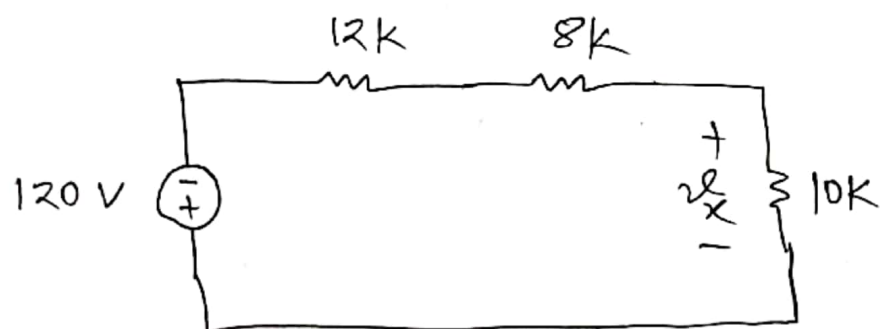
steps 4 & 5 $(16k + 1.8k + 2.2k = 20k)$; $(80V + 20k) \rightarrow (4mA \parallel 20k)$



steps 6 & 7 $(20k \parallel 30k = 12k)$; $(14mA - 4mA = 10mA)$



step 8 $(10mA \parallel 12k) \rightarrow (120V + 12k)$



step 9 (voltage division)

$$V_x = \frac{10}{12 + 8 + 10} \times (-120)$$

$$\boxed{V_x = -40V}$$